

Inorganic Chemistry

\section{Course Description}

Inorganic chemistry covers the chemistry of almost all elements in the periodic table and deals with many fundamental topics in chemistry such as bonding. This course will introduce you to the theories of chemical bonding and their application to inorganic systems with emphasis on transition metal complexes. We will explore molecular symmetry, acid/base chemistry, electrochemistry (redox chemistry), solid-state chemistry, and introduce organometallic and bioinorganic chemistry.

Inorganic chemistry interfaces with other sub-disciplines of chemistry and has found many applications in modern society, such as the renewable energy field that would help resolve barriers in reaching carbon-zero and addressing the climate crisis. As a researcher in the field of electrocatalysis for renewable energy, I will infuse my lectures with research and career opportunities in this exciting field.

Course Details:

- **Prerequisites:** Grade of C or better in the prerequisite course
- **Meeting Time:** Monday and Wednesday, 12:30 p.m-1:45 pm
- **Location:** [Building and Room Number] (SF State campus map)
- **Required Materials:**
 - Miessler and Tarr, Inorganic Chemistry (5th ed.), Prentice Hall, 2014
 - LibreTexts (open-source)
 - UCI open course resources
- **Learning Platform:** Canvas (<https://canvas.sfsu.edu>)

\section{Instructor Contact}

- **Name:** [Instructor Name] (She/Her/Hers)
- **Department:** [Department Name]
- **Email:** [Email address]
 - *When you send emails to me, please start with the course code in the title*
- **Phone:** [Phone number]
- **Office Location:** Thornton Hall [Room Number] (SF State campus map)
- **Office Hours:** By appointment in-person or via zoom
- **Research Areas:** Inorganic materials chemistry, electrochemistry for energy
- **Group Website:** [URL]

\section{Class Communications}

I welcome you to contact me outside class hours. The best way to reach me is through email - please include the course code in the subject line of all emails to ensure I see them promptly. I strive to respond to emails within 24 hours during weekdays.

For course content questions, you are encouraged to:

1. Post in our Canvas discussion forum (this helps your classmates who might have similar questions)
2. Visit me during office hours (in-person or via Zoom)
3. Speak with me before or after class for quick questions

Canvas will be our primary platform for all course materials, announcements, assignments, and grades. Please check Canvas at least once a week, as there will be weekly quizzes posted there to help you solidify important concepts.

\section{Course Objectives}

In this course, we will:

- Explore the chemistry of elements across the periodic table
- Understand chemical bonding theories and apply them to inorganic systems
- Learn about transition metal complexes and their properties
- Study molecular symmetry and group theory
- Examine acid/base chemistry and electrochemistry principles
- Investigate solid-state chemistry
- Introduce concepts in organometallic and bioinorganic chemistry
- Connect inorganic chemistry principles to real-world applications, particularly in renewable energy

\section{Teaching Methods}

My teaching philosophy is that everyone can succeed in this course, and our goals are that we are learning and improving together. I am dedicated to promoting excellence, equity, and inclusion in Inorganic Chemistry education.

I will adopt active learning approaches in my classroom, which have been shown to improve student learning outcomes. Based on my observations teaching this course in the past five years, active participation in lectures and communications with peers and the instructor provide positive impacts on students' success.

The course materials are divided into three phases to help you digest the materials and achieve the learning outcomes better. To accommodate different learning styles, readings from the

textbook, open-source materials such as LibreTexts, and watching videos from open courses are all welcome.

I highly encourage and welcome everyone to actively participate in all class activities, ask questions, and share your knowledge with your friends and family! We expect an inclusive and judgment-free learning environment where we show respect to everyone in our learning community.

\section{Course Outcomes}

Students who successfully complete this course will be able to:

- Explain periodic table trends and reactivity of elements and their compounds based on the electronic properties of atoms
- Apply symmetry operations and group theory to construct and analyze molecular vibrations
- Determine the structure of transition metal complexes through knowledge of bonding properties of various ligands as well as the electronics of the metal center
- Discuss basic bonding theories and use them to explain spectroscopic properties of transition metal complexes
- Distinguish between the various mechanisms for ligand substitution in terms of kinetics and reactivity of transition metal complexes
- Describe the basic theory behind electrochemistry and common electrochemical characterizations
- Describe the principles behind solid state chemistry in the context of thermodynamics, packing geometry, and natural abundance of inorganic compounds

\section{Grading}

To facilitate your learning, the course tasks are categorized as follows:

- **Class Participation:** 15% of final grade
 - Weekly Canvas quiz (7.5%): One-point multiple-choice question based on lecture materials
 - Group activities and literature discussion participation (7.5%): Based solely on your participation as recorded by the instructor
- **Homework Assignments:** 25% of final grade
 - Bi-weekly homework assigned on Canvas (6-7 assignments total)
 - Each homework includes several free-response questions (1-2 hours to complete)
 - Submit as a PDF file on Canvas for instructor's grading and feedback

- Late homework submission before the release of solutions is eligible for 50% partial credit
- If you need an extension, please let me know in advance
- **Exams:** 60% of final grade
 - Four in-class exams (no cumulative final)
 - You can bring an A4-size cheat sheet with information you need
 - The best 3 exam grades out of 4 will be counted toward the final grade
 - Graded exams will be returned with comments and feedback
 - You are welcome to discuss comments and feedback during office hours

Exam Schedule:

- Exam 1: 2/28 (75 min)
- Exam 2: 3/20 (75 min)
- Exam 3: 4/24 (75 min)
- Exam 4: 5/22 (120 min)

Grading Scale:

- A: 90% and above
- A-: 87-89%
- B+: 84-86%
- B: 80-83%
- B-: 77-79%
- C+: 74-76%
- C: 70-73%
- C-: 67-69%
- D: 60-66%
- F: Below 60%

\section{Attendance Policy}

You are expected to attend every class, but I understand that extenuating circumstances arise that can make this difficult. If you cannot attend a class, please let me know via email in advance.

I strongly encourage your active participation in this course. I've found this is the best way to engage you in learning the materials, and it is most effective to learn in the classroom with your peers instead of figuring out the materials alone by yourself.

Important dates to note:

- First day of instruction: 1/29
- Instructor drop deadline: 2/15

- Last day to add/drop without a "W": 2/16
- Spring recess (no classes): 3/25 - 3/29
- No class: 4/1
- Last day of instruction: 5/15

\section{DPRC}

Students with disabilities who need reasonable accommodations are encouraged to contact the instructor. The Disability Programs and Resource Center (DPRC) is available to facilitate the reasonable accommodations process. The DPRC is located in the Student Service Building and can be reached by telephone (voice: 415-338-2472, video phone: 415-335-7210) or by email (dprc@sfsu.edu).

Academic Integrity Policy:

Academic integrity, the ethical presentation of one's own work in accordance with the rules established for this class, is required. Instances of academic misconduct will be reported to the College in which the course is housed, the Division of Graduate Studies (if a graduate student), and the Office of Student Conduct.

Although I encourage discussions with your classmates, I am strongly against directly copying other people's work, including from your classmates or sources on the internet. If similar work on homework or exams is identified, this would be considered plagiarism and lead to zero credit for this assignment for both parties.

Course Materials Sharing Policy:

I highly encourage taking notes during class instead of taking photos of the notes. You may not capture audio, photos, or video from class sessions on your own devices without explicit permission of the instructor and everyone present, unless part of a DPRC-authorized accommodation. You should not post any course materials to third-party sites (such as Chegg) or post any recordings, screenshots, audio, or chat transcripts in any setting outside the class. Violations of this are subject to student disciplinary action.

Student Disclosures of Sexual Violence:

SF State fosters a campus free of sexual violence including sexual harassment, domestic violence, dating violence, stalking, and/or any form of sex or gender discrimination. If you disclose a personal experience as an SF State student, the course instructor is required to notify the Title IX Coordinator. To disclose any such violence confidentially, contact:

- The SAFE Place - (415) 338-2208; http://www.sfsu.edu/~safe_plc/
- Counseling and Psychological Services Center - (415) 338-2208; <http://psyservs.sfsu.edu/>

